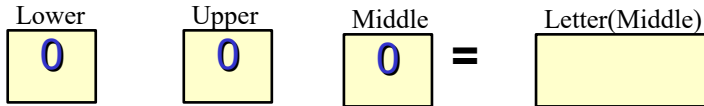


**A binary search
to find the letter “P”
in the letter table**

LetterTable

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26



```
MOVE 1 TO Lower
MOVE 26 TO Upper
PERFORM UNTIL ItemFound OR ItemNotInTable
  COMPUTE Middle = (Lower + Upper) / 2
  EVALUATE TRUE
    WHEN Letter(Middle) < "P" COMPUTE Lower = Middle + 1
    WHEN Letter(Middle) > "P" COMPUTE Upper = Middle - 1
    WHEN Letter(Middle) = "P" SET ItemFound TO TRUE
    WHEN Lower > Upper THEN SET ItemNotInTable TO TRUE
  END-EVALUATE
END-PERFORM
```

LetterTable

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower	Upper	Middle	=	Letter(Middle)
1	26	0		

MOVE 1 TO Lower

MOVE 26 TO Upper

PERFORM UNTIL ItemFound OR ItemNotInTable

 COMPUTE Middle = (Lower + Upper) / 2

 EVALUATE TRUE

 WHEN Letter(Middle) < "P" COMPUTE Lower = Middle + 1

 WHEN Letter(Middle) > "P" COMPUTE Upper = Middle - 1

 WHEN Letter(Middle) = "P" SET ItemFound TO TRUE

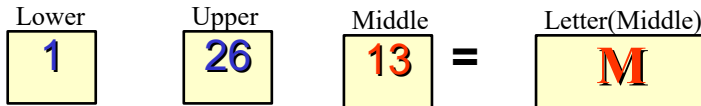
 WHEN Lower > Upper THEN SET ItemNotInTable TO TRUE

 END-EVALUATE

END-PERFORM

LetterTable

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26



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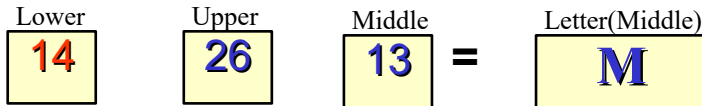
WHEN Lower > Upper THEN SET ItemNotInTable TO TRUE

END-EVALUATE

END-PERFORM

LetterTable

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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Lower	Upper	Middle	=	Letter(Middle)
14	26	20	=	T

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MOVE 26 TO Upper

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WHEN Letter(Middle) = "P" SET ItemFound TO TRUE

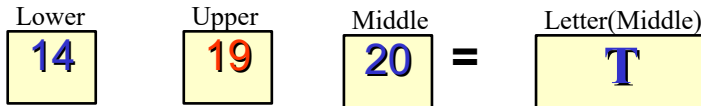
WHEN Lower > Upper THEN SET ItemNotInTable TO TRUE

END-EVALUATE

END-PERFORM

LetterTable

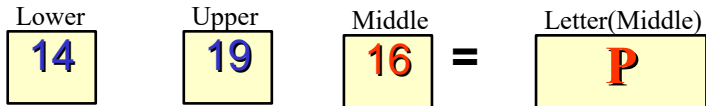
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26



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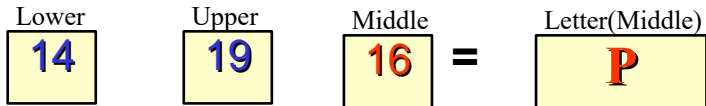
WHEN Lower > Upper THEN SET ItemNotInTable TO TRUE

END-EVALUATE

END-PERFORM

LetterTable

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